

Inventory Forecasting

Read-only demand prediction powered by BuyAbans data

01

CONNECT

Authenticated BuyAbans API

02

CURATE

Sync, normalize and reconcile

03

PREDICT

Dense series + four algorithms

04

DECIDE

Explainable human review

Read-only business data

All 3 demand grains synced

EWMA remains the default

The system now has a clear job

Convert back-office data into management visibility and demand forecasts without creating a second operational truth.

1

READ

Catalog, current stock and order history arrive through authenticated GET endpoints.

2

TRUST

Sync is idempotent; variants and attributes are normalized; demand grains reconcile.

3

PREDICT

Daily series are densified, forecasted with four algorithms and shown with uncertainty.

4

GOVERN

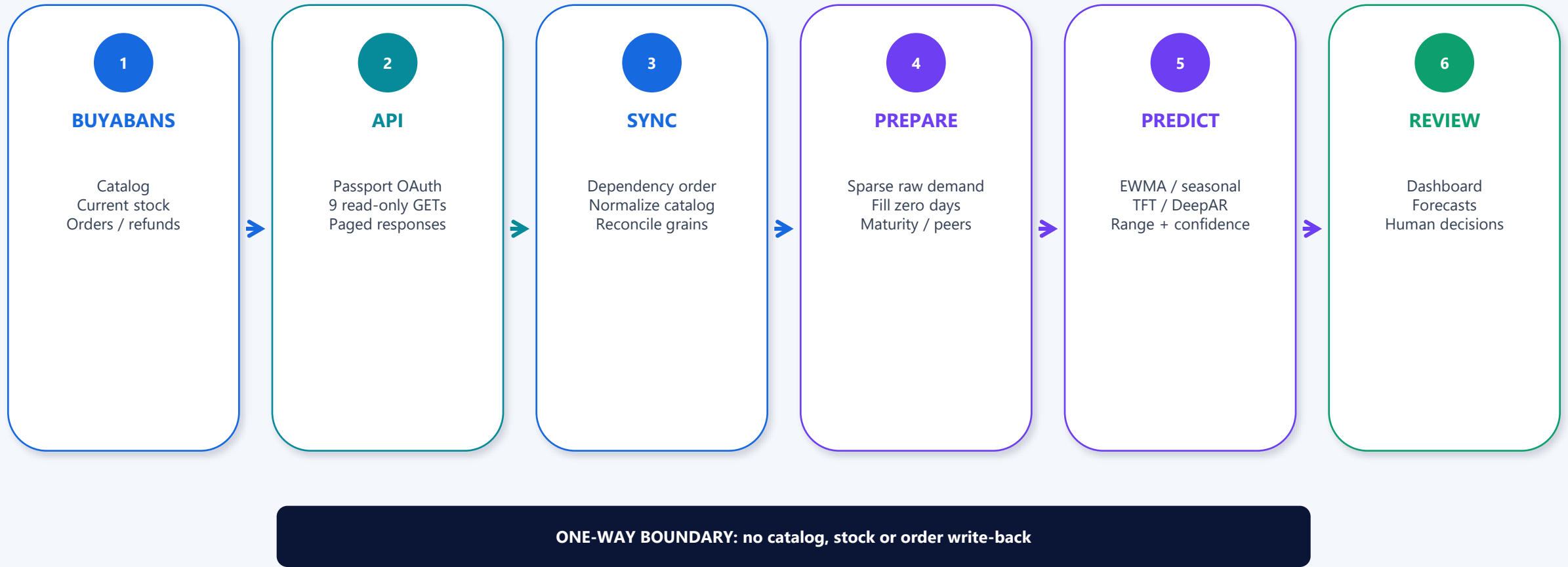
EWMA remains default; decisions stay human; nothing writes back to BuyAbans.

Management boundary

The forecast path is current. Recommendations and several legacy analytics still need a BuyAbans stock-to-location design before they can be trusted as production signals.

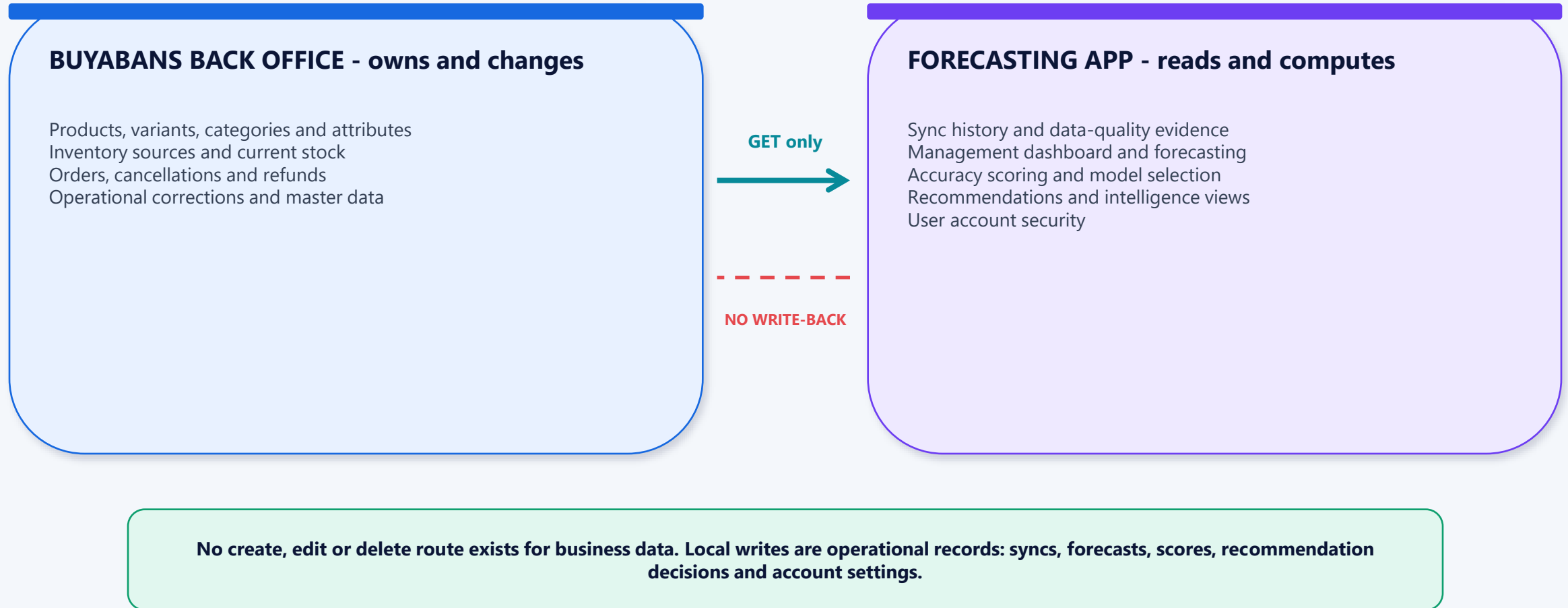
End-to-end: source data to human decision

A one-way data path protects the back office while creating an analytical and forecasting layer.



One system of record; one prediction layer

The separation prevents local edits from being overwritten or mistaken for operational truth.



What users can see and do today

The sidebar is organized around visibility, forecasting and investigation - not stock operation.

OVERVIEW

Dashboard
BuyAbans sync

CATALOG

Products, variants, SKUs
Categories, brands, attributes

INVENTORY VIEWS

Current stock, analytics, batches
Snapshots and warehouses

SUPPLY REFERENCE

Read-only suppliers
Historical operations remain by URL

FORECASTING

Forecasts and runs
Recommendations and allocation

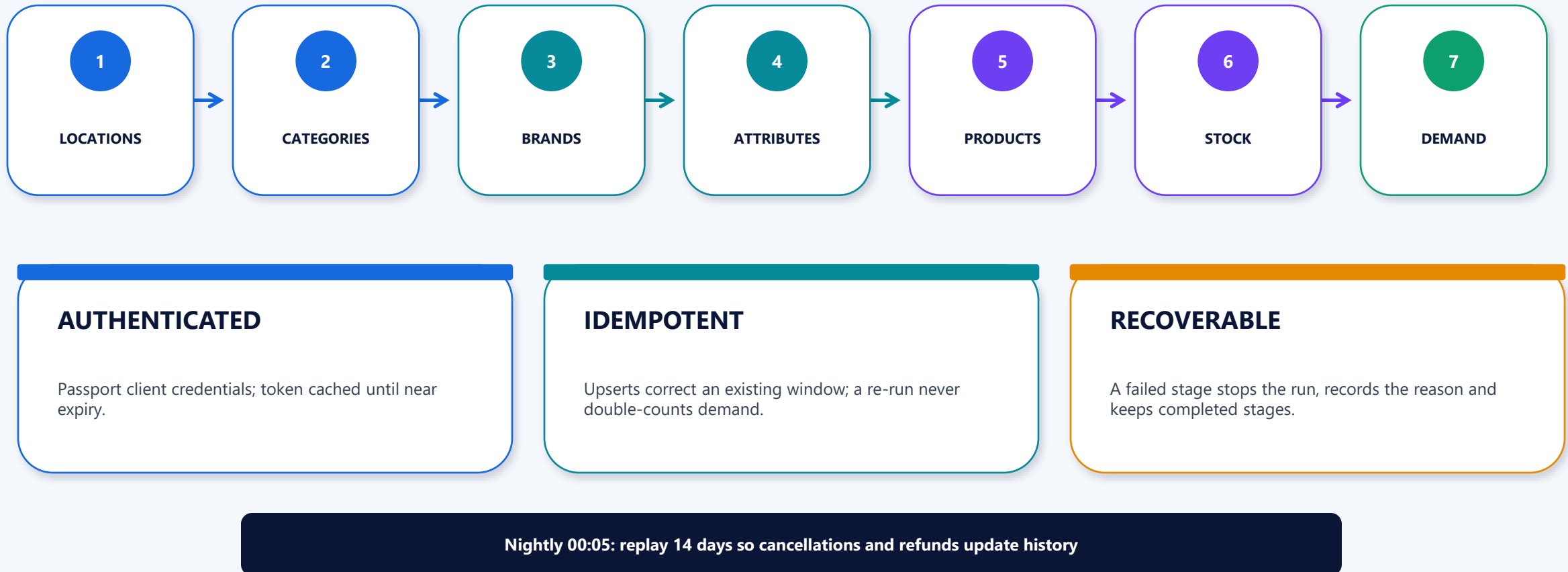
ADVANCED INTELLIGENCE

Supplier, demand, product, price
and promotion signals

All pages require authentication + verified email; role-based permissions are not implemented.

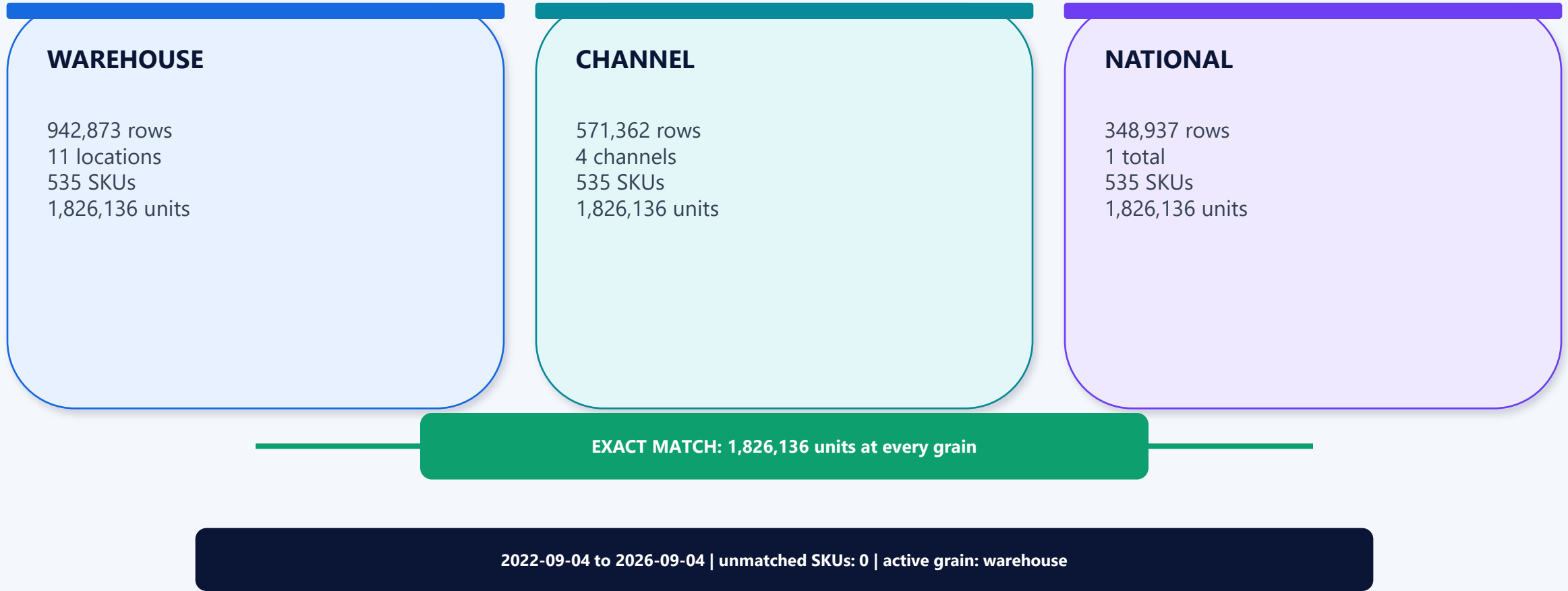
How a BuyAbans sync becomes usable data

Dependencies run in order, re-runs correct existing rows, and failures remain visible in sync history.



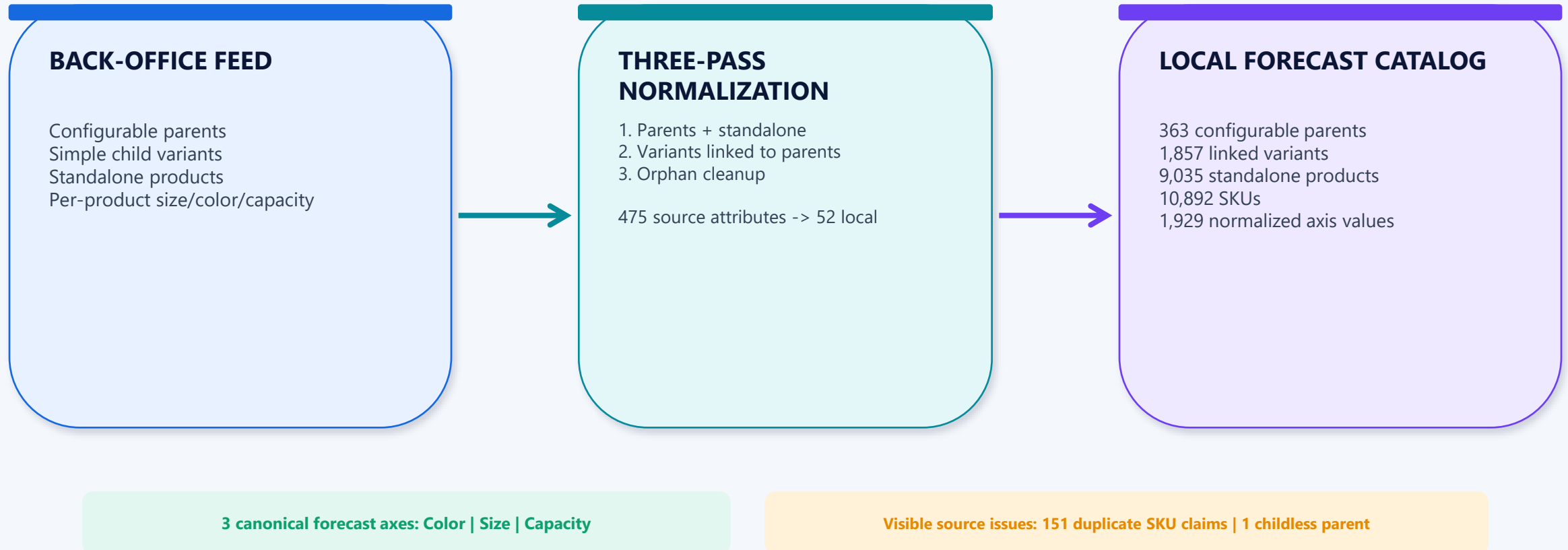
All three demand views reconcile to the unit

Different grains describe the same sales from different angles; they must never be summed together.



The sync rebuilds a usable product tree

Parentage and common variant axes are normalized during ingestion instead of flattening or duplicating the catalog.



A missing row means zero sales - not a missing day

The source feed stays sparse; every forecasting consumer reconstructs the calendar before modelling.



Before correction



51,919 forecast

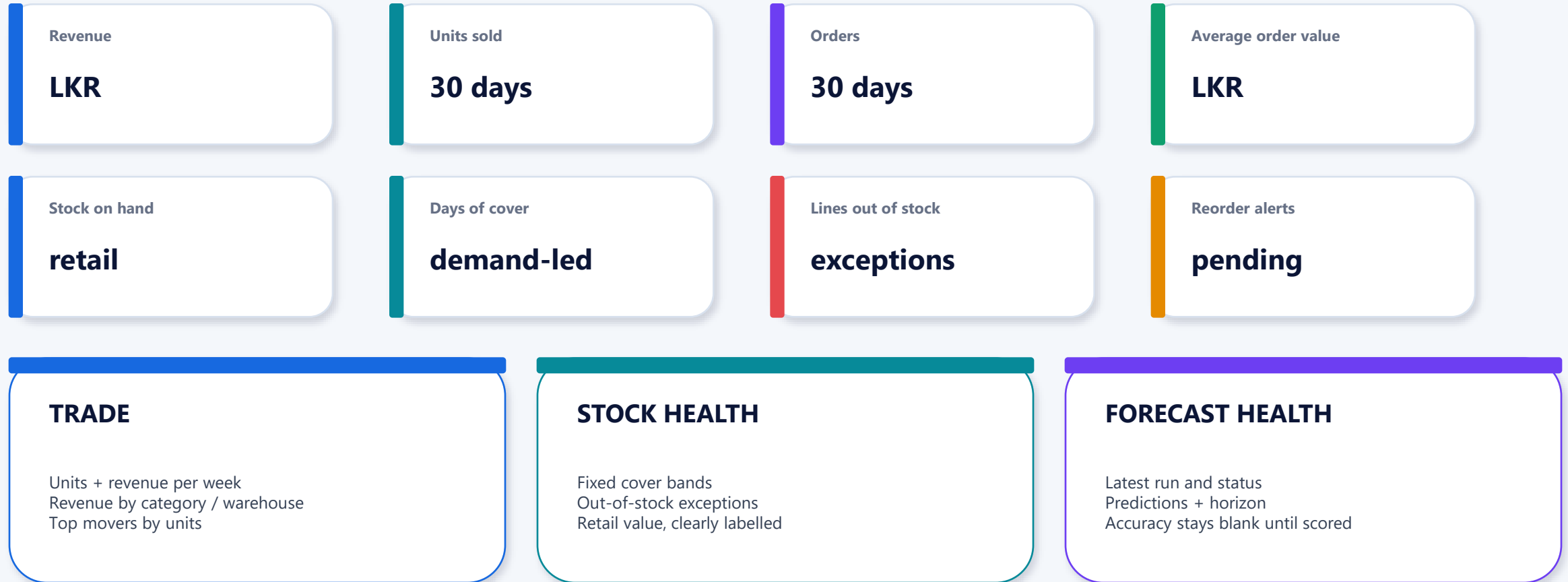
After correction



27,952 vs 27,132 actual - within 3%

The dashboard is now a management briefing

Every window ends on the last synced demand date, so stale data never masquerades as a run of zero sales.



The forecast page answers: what happens next?

Plain language, a comparison window, a likely range and visible confidence replace technical codes.

EXPECTED UNITS

31,365

+3% versus the preceding 30 days

PREVIOUS WINDOW

30,445

Observed sales in the same-length window

LIKELY RANGE

19,314-44,287

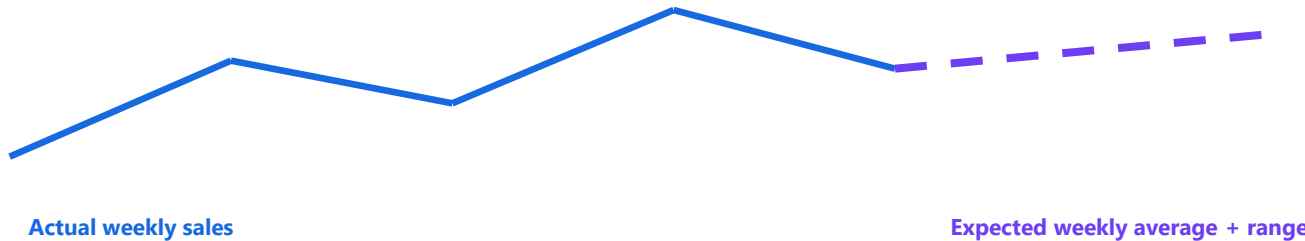
Use the range when confidence is low

COVERAGE

533 products

Overall confidence: Low

SALES SO FAR, AND WHAT COMES NEXT



THE TABLE SPEAKS BUSINESS

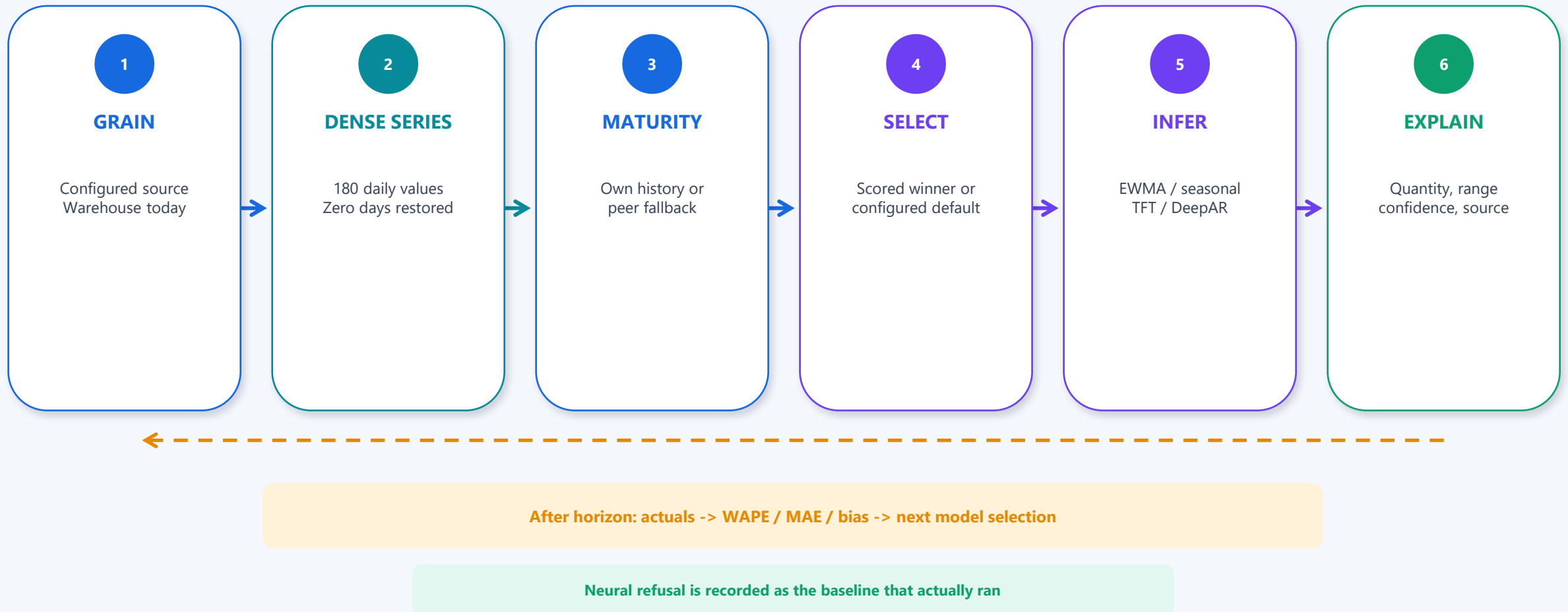
Expected to sell
Over the next N days
How sure
Based on
How it turned out

Latest run by default

Generated-data evidence - not production demand

How one forecast is produced and governed

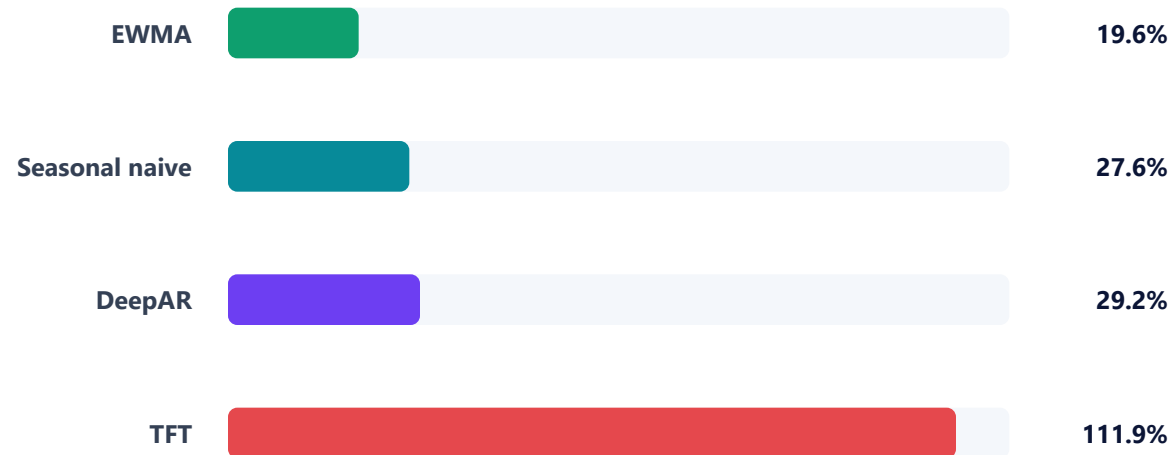
Selection happens before inference; scoring happens only after the predicted horizon has finished.



EWMA remains the evidence-based default

Lower WAPE is better. Saved evaluations are from generated BuyAbans history and prove the pipeline, not real-world accuracy.

LATEST SAVED 30-DAY CROSS-MODEL EVALUATION



Actual: 26,592 units | EWMA predicted: 28,753

NEURAL TRAINING ARTIFACTS

365-day holdout
TFT WAPE 98.3% | bias -2.4%
DeepAR WAPE 102.0% | bias +12.6%

Checkpoints + dataset parameters regenerated on 04 Sep.

Do not switch without real scored history.

CONFIGURED DEFAULT: EWMA

The widened dataset is usable at operating scale

Indexes, date-range corrections and worker limits were measured against the current four-year dataset.

SYNCED DEMAND

1.826M units

Each grain reconciles to the same total

FORECAST RUN

2,300 rows

Completed in 2m22s across 2,259 request pairs

DASHBOARD

2.5s

Reduced from 70.4s

FORECAST PAGE

0.9s

Reduced from 19.5s

WHY THE INDEXES MATTER

Back office: orders (created_at, status)

App: daily demands (grain, sku_id)

Distinct-SKU query: 8.92s -> 0.44s

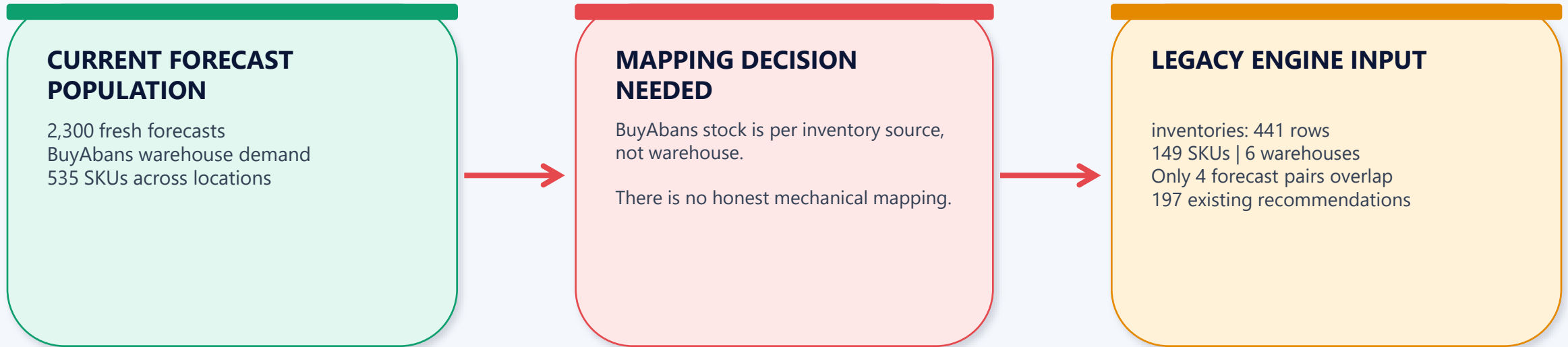
WHY THE WORKER LIMIT MATTERS

A 60-second listener silently killed a full run and left it at Processing.

Job + listener timeout: 1,800 seconds. Production needs stale-run alerts.

Recommendation workflow exists; its stock input is not current

Forecasts and current BuyAbans stock do not yet meet at a compatible location grain.



Safe claim today

Accept / Modify / Reject is implemented and auditable. Present quantities and the reorder-alert count are legacy-data demonstrations. Nothing auto-creates a purchase order or transfer.

Not every screen is on the same freshness path yet

Production trust depends on knowing exactly which source each view reads.

CURRENT BUYABANS PATH

- Dashboard trading metrics
- Revenue/category/location charts
- Forecast training and serving
- Maturity and peer profiles
- Catalog, variants, attributes and stock

LEGACY OR FROZEN INPUT PATH

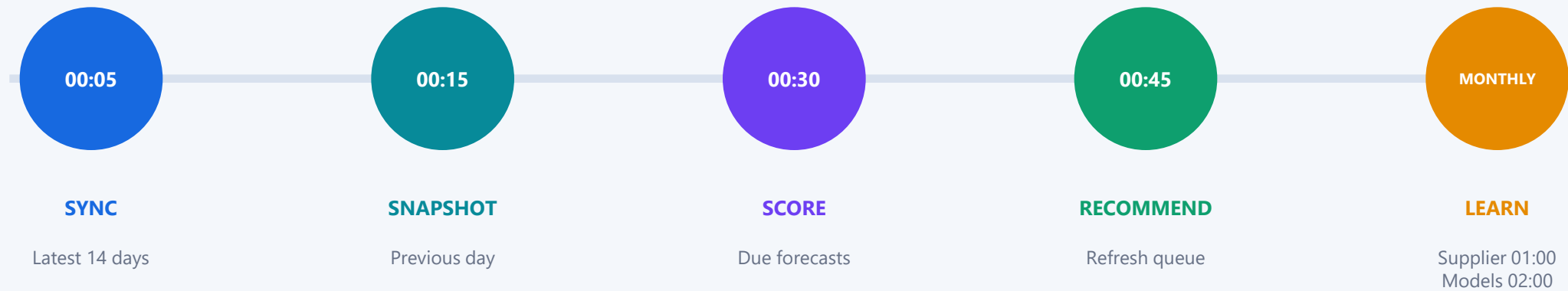
- Inventory analytics and recommendations
- Lost sales and anomalies
- Supplier performance
- Product relationships / elasticity
- Promotion impact

Functional, but not replenished by this sync.

Demonstrate capability, state lineage, never imply all figures share the live sync.

The daily and monthly operating cycle

Sequence protects dependency: synchronize first, then capture, score and recommend.

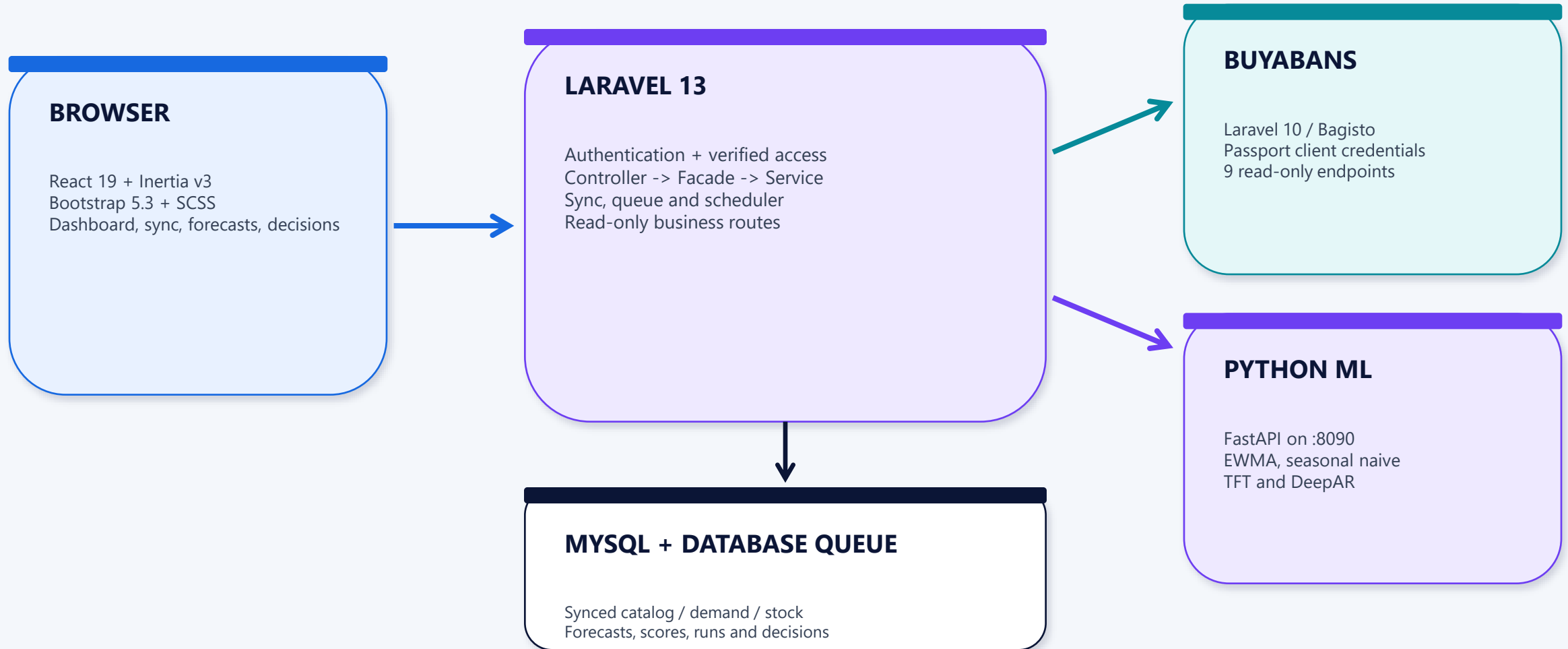


Operating requirement

Host cron + supervised queue + supervised Python + alerts. Snapshot/recommendation jobs still follow the legacy inventory path until stock mapping is resolved.

Two applications, two runtimes, one analytical experience

Laravel owns orchestration; Python provides inference; BuyAbans remains external truth.



Strong boundaries are delivered; authorization is not

Authentication, data integrity and model honesty exist today; production access governance still needs ownership.

ACCOUNT CONTROLS

- Email verification
- Password reset + confirmation
- TOTP two-factor authentication
- Paskeys
- Login/challenge throttling

DATA CONTROLS

- Read-only business routes
- OAuth machine identity
- Idempotent sync + run history
- Grain isolation / reconciliation
- Named neural fallback
- Human decision record

REQUIRED FOR PRODUCTION

- Role-based permissions
- Controlled registration
- Real mail + cookie policy
- Secret rotation / token protection
- Backups and monitoring
- Release validation

Today: every authenticated, verified user has broad visibility and operational triggers.

What has been verified in the current system

Counts show breadth; reconciliations and measured runs show integration. Neither proves adoption or real-demand accuracy.

LATEST LOGGED PEST SUITE

253 / 253

Application behavior + grain isolation

PYTHON TESTS

62 / 62

Last fully logged service suite

SOURCE BREADTH

36 models

43 migrations | 34 services | 33 facades

USER EXPERIENCE

45 React pages

Responsive Bootstrap + dark mode

INTEGRATION EVIDENCE

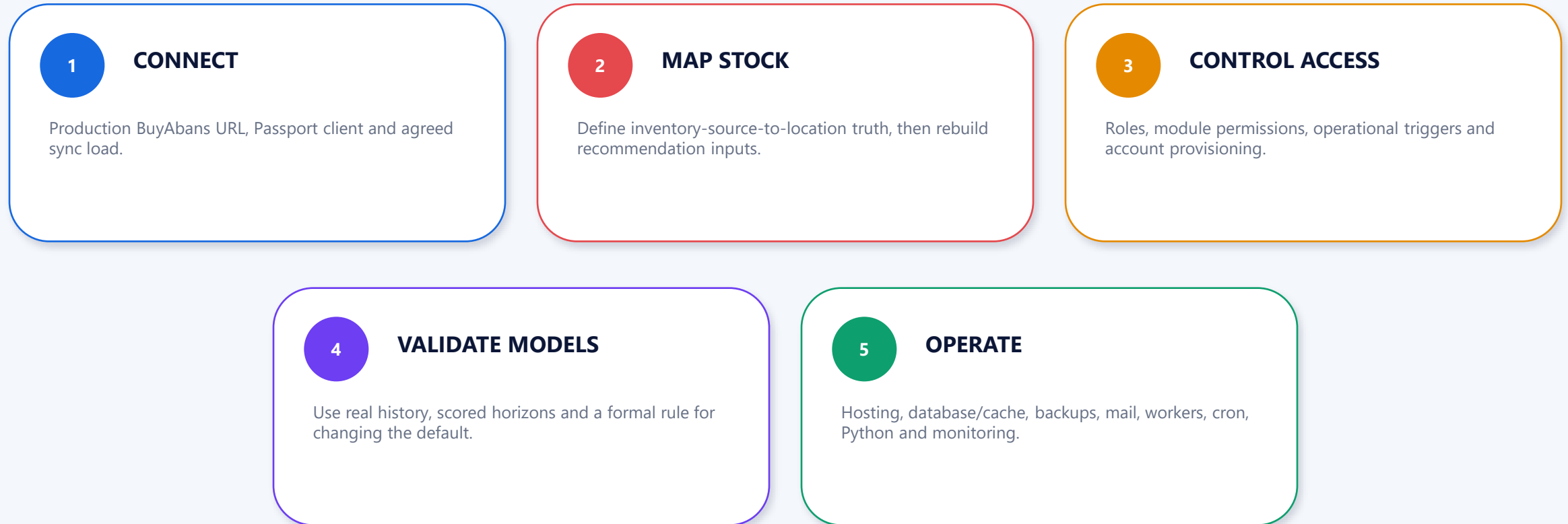
All 3 grains exactly reconciled
0 unmatched demand SKUs
2,300 forecasts in 2m22s
Dashboard and forecast performance measured

EVIDENCE BOUNDARY

Generated demand history
Training orchestration not tested in CI
Recommendations read legacy inventory
Roles and production operations unverified

Five decisions turn the demo into a production service

The remaining work is data ownership, access policy, validation and reliable operation.



CURRENT POSITION: management demo / controlled pilot - not production decision automation

A clear eight-minute route

1

DASHBOARD

Trade, stock position and forecast health

2

BUYABANS SYNC

Connection, stages and run history

3

PRODUCTS + VARIANTS

Tree, normalized axes and SKU identity

4

FORECASTS

Headline, range, confidence and basis

5

FORECAST RUNS

Queued processing and completed evidence

6

RECOMMENDATIONS

Show workflow - state the stock-source gap

Close with the accurate story:

BuyAbans remains operational truth. This application turns that truth into visible performance, explainable forecasts and governed decisions.